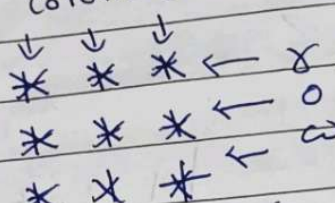


Week 1 Module - C++

1. star Patterns (inner loops)

① Square → 

logic: outer loop (rows)
3 times or n-times
for (i=0; i<3; i++)
or
for (i=0; i<n; i++)

inner loops (columns)

for (j=0; j<3; j++)
or
for (j=0; j<n; j++)

Code: #include <iostream>

using namespace std;

(++int main;

{

(++int h;

cout >> "Enter the size of the square";

(cin >> h;

for (int i=0; i<h; i++)

{

for (int j=0; j<n; j++)

```

    {
        cout << endl; "*" ;
    }
    cout << endl;
}
return 0;
}

```

②

Hollow square →

	1	2	3	4	5
1	*	*	*	*	*
2	*				*
3	*				*
4	*				*
5	*	*	*	*	*

Logic:

if $i == 0$
 $i == 5$

Print "*" "

$j = 1$

$j = 5$

else

cout << " - ";

Code: for (int i = 0; i < n; i++)

{

for (int j = 0; j < n; j++)

{

if (i == 0 || i == n - 1 || j == 0 || j == n - 1)

{

cout << "*" ;

else

else
 { cout << " ";

}
 cout << endl;

③

Triangle

→

1	*				
2	*	*			
3	*	*	*		
4	*	*	*	*	
5	*	*	*	*	*

Logic → so, outer loop are connected by each other like

1 row only 1 star in column

2 " " 2 " " " "

and so on

Code: for(int i=0; i<n; i++)

{

for(int j=0; j<=i; j++)

{

cout << " * ";

}

cout << endl;

}

}

④

Inverted Triangle

→

* * *

* *

*

Logic → Same as upax bas

ise bas connect

n-1

Code →

```

for (int i = 0; i < n; i++)
{
    for (int j = 0; j < n - i; j++)
    {
        cout << " * ";
    }
    cout << endl;
}
    
```

1 2 3 4 5 6 7

5) Pyramid →

```

1      *
2     * * *
3    * * * * *
4   * * * * * *
    
```

Logic →

```

outer loop
for (int i = 0; i < n; i++)
    for space
        for (int j = 0; j < n - i - 1; j++)
    
```

inner loop

```

for (int j = 0; j < 2 * i + 1; j++)
    
```

Code →

```

for (int i = 0; i < n; i++)
{
    for (int j = 0; j < n - i - 1; j++)
    {
        cout << " ";
    }
    for (int j = 0; j < 2 * i + 1; j++)
    {
        cout << " * ";
    }
    cout << endl;
}
    
```

⑥ Diamond →

```
  *
 * * *
* * * *
 * * *
  *
```

Logic: Pyramid + Inverted Pyramid

Code: - for (int i = 0; i < n; i++)

{

for (int j = 0; j < n - i - 1; j++)

{

cout << " ";

}

for (int j = 0; j < 2 * i + 1; j++)

{

cout << " * ";

}

cout << endl;

}

for (int i = n - 2; i >= 0; i--)

{

for (int j = 0; j < n - i - 1; j++)

{

cout << " ";

}

for (int j = 0; j < 2 * i + 1; j++)

{

cout << " * ";

}

cout << endl;

}

⑦ Half Diamond →

```

*
**
***
****
*****
****
***
**
*
  
```

logic → upper half

no of rows = no of star to print

lower half

no. of rows = no of star to print --

Code →

```

for (int i = 1; i <= n; i++) {
    for (int j = 1; j <= i; j++) {
        cout << " ";
    }
    cout << endl;
}
  
```

```

    cout << endl;
  
```

```

for (int i = n - 1; i >= 1; i--) {
    for (int j = 1; j <= i; j++) {
        cout << " ";
    }
    cout << endl;
}
  
```

```

    cout << endl;
  
```

```

}
  
```

```

return 0;
  
```

```

}
  
```

⑧ Half Diamond Inverted

```

* * * * *
 * * * *
  * * *
   * *
    *
  
```

logic → Inverted Pyramid

```

code → for (int i = 0; i < n; i++) {
        for (int j = 0; j < i; j++) {
            cout << " ";
        }
        for (int j = 0; j < 2 * (n - i) - 1; j++) {
            cout << " * ";
        }
        cout << endl;
    }
}
  
```

#2 Number Pattern

1. Square →

```

1 1 1 1 1
1 1 1 1 1
1 1 1 1 1
  
```

Same as Previous just cout << "1"

② Incrementing

```

1 2 3 4
1 2 3 4
1 2 3 4
1 2 3 4
    
```

logic → Outer loop

code for(int i=1; i<=n; i++) {

// inner loops

for(int j=1; j<=n; j++)
 & {

1- (i) code → cout << j << " ";

cout endl;

}

③ Right angle Triangle →

```

1 2
    
```

```

1 2 3
    
```

```

1 2 3
    
```

Same as Previous Right Triangle
 but cout << j << " ";

④ Half diamond inverted pattern just previous one.